FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II

PAPER-1

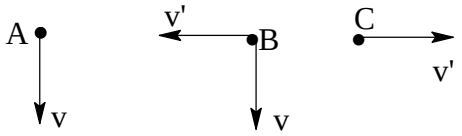
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**ABC**	21.	**BC**	41.	**ABCD**
2.	**ABD**	22.	**ABD**	42.	**AC**
3.	**ABC**	23.	**ABCD**	43.	**BCD**
4.	**ABD**	24.	**ABCD**	44.	**D**
5.	**BD**	25.	**ABC**	45.	**CD**
6.	**ACD**	26.	**AB**	46.	**ACD**
7.	**ACD**	27.	**AD**	47.	**A**
8.	**AC**	28.	**AD**	48.	**B**
9.	**AC**	29.	**ABCD**	49.	**ABC**
10.	**ACD**	30.	**ABCD**	50.	**AC**
11.	**9**	31.	**6**	51.	**6**
12.	**5**	32.	**5**	52.	**1**
13.	**0**	33.	**4**	53.	**4**
14.	**4**	34.	**8**	54.	**3**
15.	**0**	35.	**6**	55.	**4**
16.	**5**	36.	**4**	56.	**4**
17.	**2**	37.	**7**	57.	**1**
18.	**3**	38.	**8**	58.	**6**
19.	**2**	39.	**6**	59.	**1**
20.	**8**	40.	**7**	60.	**3**

Physics

PART – I

SECTION – A

1. ABC
   Spring can be in the state of extension, compression or natural length in equilibrium
2. ABD
3. ABC
4. ABD
5. BD
6. ACD
7. ACD
   Velocity direction of A, B and C



$$\frac{1}{2}mv^2+\frac{1}{2}m\left(v^2+\frac{v^2}{4}\right)+\frac{1}{2}\frac{mv^2}{4}=2mgh$$

$$v=\sqrt{\frac{8}{5}gh}=2\sqrt{2}m/s$$

8. AC
9. AC
10. ACD

SECTION – C

11. [9]
    Field inside the shell is zero, so potential is uniform

$$V=\frac{kq}{2R}=9\,\text{kilo volts}$$

12. [5]

$$\frac{-3}{5}(50)x-\left(\frac{1}{2}\times 50x^2\right)=0-\frac{1}{2}\times 5\left(\frac{4}{5}\sqrt{10}\right)^2$$

$$25x^2+30x-16=0 \Rightarrow x=\frac{2}{5}\,\text{meters}$$

So at maximum compression spring force $=50\left(\frac{2}{5}\right)=20N$ and friction force $=\frac{3}{5}(50)=30N$.

Hence the block will not move so distance $=\frac{2}{5}m$.

13. [0]
14. [4]
15. [0]
16. [5]
17. [2]

$$t(\mu-1)=\lambda \quad \Rightarrow \quad t(1.5-1)=\lambda \quad \Rightarrow t=2\lambda$$

18. [3]
Energy required to remove the electron from the third orbit
$= \frac{13.6}{3^2}\text{eV} = 1.5 \text{ eV} = 1.5 \times 1.6 \times 10^{-19}\text{J} = 3 \text{ eV}$

19. [2]
$$\frac{f}{-u_1 + f} = -\frac{f}{(-u_2 + f)} \Rightarrow -u_2 + f = u_1 - f \Rightarrow f = \frac{u_1 + u_2}{2} = 2$$

20. [8]
$$d = \frac{3.6}{1.2} + \frac{7}{1.4} = 8$$

Chemistry

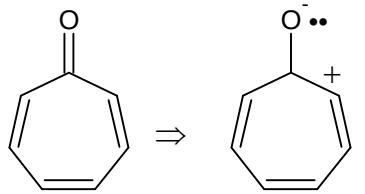
PART – II

SECTION – A

21. BC

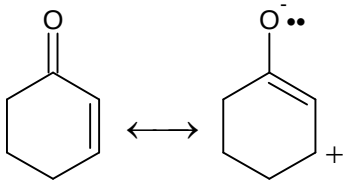
(A)

(B)

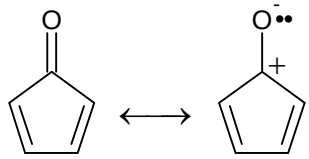


aromatic

(C)



(D)



anti aromatic

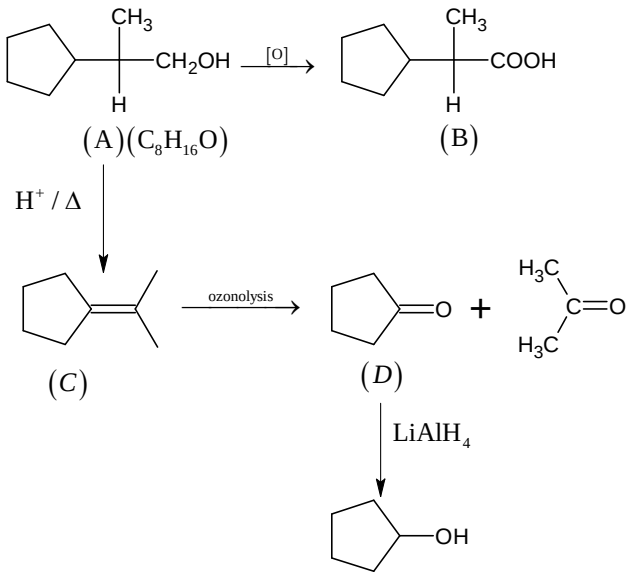
22. ABD

Facts

23. ABCD

Facts

24. ABCD



25. ABC

A lone pair of N-atom participating in resonance will be less basic

26. AB

Reaction proceeds by limiting reagent only. Enthalpy of neutralization decreases, when acid or base or both are weak:

27. AD

Chiral molecules do not have plane of symmetry

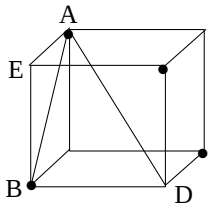
28. AD

BaO_2 and Na_2O_2 are peroxides

29. ABCD
30. ABCD

SECTION – C

31. [6]



Face diagonal of cube $(AB)=2R$

Edge length of cube $(BE)=\frac{2R}{\sqrt{2}}$

Body diagonal of cube $(AD)=\sqrt{3}\times\frac{2R}{\sqrt{2}}$

$=\sqrt{6}\times R$

32. [5]

$Mg^{2+}+2e^{-}\longrightarrow Mg\left(E_{Mg}=\frac{24}{2}=12g\right)$

Charge required for the deposition of $12g\ Mg=1F$

Charge required for the deposition of $1g\ Mg=\frac{1}{12}F$

$\therefore$ cost for $\frac{F}{12}$charge=Rs.5

$Al^{3+}+3e^{-}\longrightarrow Al\left(E_{Al}=\frac{27}{3}=9g\right)$

Charge required for $9g=1F$

Charge required for $30g=\frac{30}{9}F=\frac{10F}{3}$

Cost for $\frac{10F}{3}$charge$=5\times12\times\frac{10F}{3}$

$=Rs.200$

33. [4]

$pKa\ of\ HX=14-10=4$

$pH=pK_a+\log\frac{1}{1}=4$

34. [8]

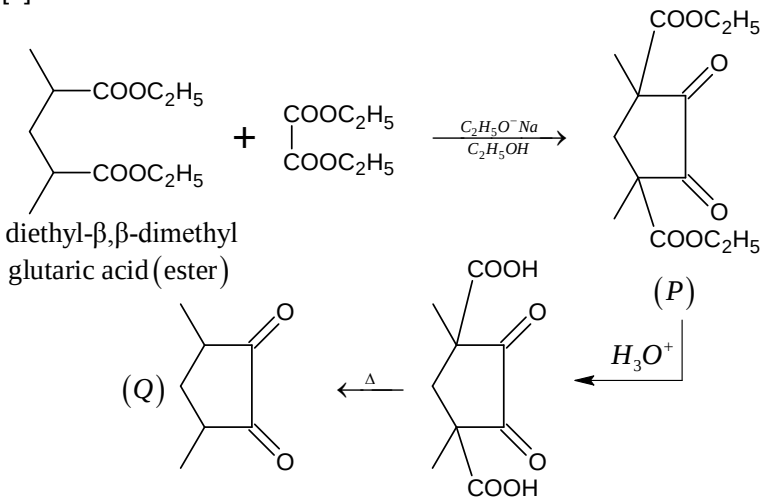
$(CO)_{x/2}\,Co-Co(CO)_{x/2}$

$EAN\,(=36)=27+1+2\times\frac{x}{2}$

$\Rightarrow 28+x=36$

$x=8$

35. [6]



36. [4]

$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g) \rightleftharpoons HCl(g)\,\Delta G^\circ = 1.72\,kJ\,mol^{-1}$

ΔG° for $2HCl(g) \rightleftharpoons H_2(g)+Cl_2(g)$ is $\left[2\times(-1.72)\right]kJ$

$\Delta G^\circ = -2.303RT\log K_p$

$-3.44\times10^3 = -5700\log K_p$

$\log K_p = 0.6$

$K_p = 4$

37. [7]

Salt of $WA \& WB$

$pH = 7+\frac{1}{2}(pK_a - pK_b)$

38. [8]

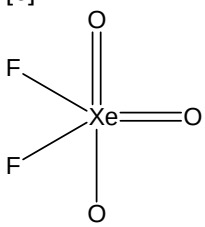
For salt of weak acid and strong base

$pH = 7+\frac{1}{2}[pK_a, \log C],$

$= 7+\frac{1}{2}\left[\left(-\log 1\times10^{-4}\right)+\log(0.01)\right]$

$= 7+\frac{1}{2}(4-2) = 8$

39. [6]



40. [7]

Green vitriol is $FeSO_4.7H_2O$

Mathematics

PART – III

SECTION – A

41. ABCD

$$f(x)=\begin{cases} px+\lambda, x>0 \\ q, x=0 \\ e^x, x<0 \end{cases}$$

42. AC

$f(x)=(2x-1)(2x+1)(x-2)(x-4)$

43. BCD

$p(n)=$ Probability that child stops after drawing exactly n marbles

$$p(n)=\frac{1}{2}.\frac{2}{3}.\frac{3}{4}.....\left(\frac{n-1}{n}\right)\left(\frac{1}{n+1}\right)=\frac{1}{n(n+1)}$$

44. D

$$\left(a^2-b^2\right)x^4-2ha^2\left(a^2-b^2\right)x^3-x^2+2a^4h\left(a^2-b^2\right)-a^6h^2=0\Rightarrow\left(\sum x_1\right)\left(\sum\frac{1}{x_1}\right)=4$$

45. CD

If α is real root then $\alpha=2, m=\pm 2,$ non-real root = $1+i$

46. ACD

If $0<x<1, a=6;$ if $x>1, a=2,$ if $x<-1, a=-2$

47. A

$$\frac{3f''(x)}{f'(x)}=\frac{f'''(x)}{f''(x)}$$

$$\Rightarrow 3\ln f'(x)=\ln f''(x)+k$$

$$\Rightarrow\frac{d^2x}{dy^2}=c$$

48. B

$$e^{-\sqrt{-\ln t}}=t^{\frac{1}{\sqrt{-\ln t}}}, t\in(0,1)$$

49. ABC

$$2S=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left(\frac{m.n}{m.3^n+n.3^n}\right)\left(\frac{n}{3^m}+\frac{m}{3^n}\right)$$

50. AC

SECTION – C

51. [6]

$p(x)=\left(x^2-4\right)\left(x^4-10x^2+10\right)$

52. [1]
53. [4]
54. [3]

55. [4]
56. [4]
57. [1]
58. [6]

$6+4-4$

59. [1]

$x_1 = 4, x_2 = 8,\ k = 3,\ w = 4$

60. [3]

$\Rightarrow \left(x^2+1\right)\left(x^2-bx+1\right) > 0$

So $b^2 - 4 < 0 \Rightarrow b \in (-2, 2)$